Generation as Computation

Generalised Computation in the Set-theoretic Universe and Beyond
(Thesis Proposal)

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Outline

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1. Introduction
   - Background: Infinitary Computation
   - Background: Formulas as Programs
   - Overview
2. Generalised Computation Within V
   - Sequential Algorithms
   - Generalised Sequential Algorithms
   - Relative Computability
3. Generalised Computation Beyond V
   - Degrees of Small Extensions
   - Local Method Definitions

Gödel's Informal Proposition

- In an essay Gödel wrote in 1958 (but only published posthumously) [4], he criticised Hilbert's choice of "intuitively clear" notions meant to make up the bedrock of classical mathematics.
- He lamented that "the domain of 'finitary mathematics"' is unnaturally restrictive.
- In the same work, Gödel informally described a notion of computation built inductively from natural numbers, a notion of which power goes beyond conventional finitary computation.
- He called this notion *computable functions of finite type*.

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More Formal and Explicit Conceptions

- The eponymous Blum-Shub-Smale machines (BSSMs) [5], born from the study of dynamical systems.
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- "A program is a formula (of a particular form)" — an oft-repeated slogan.
- An oracle machine can functionally be represented as a sufficiently simple parametrised formula in the language of arithmetic.
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- α-recursion theory, born from ideas by Takeuti [1], Kreisel and Sacks [2] and some others, in the 1960s.
- E-recursion theory, formulated by Normann [3] in the 1970s.
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In Greater Generality...

- Relative computability is but a form of parametrised definability over some structure.
- Vanilla computability is a special case whereby the parameter in question contains the least amount of information (e.g. using the empty set as parameter).
- From a set-theoretic perspective, computation can thus be thought of as procedural set (even class) generation over a possibly enriched universe of set theory.

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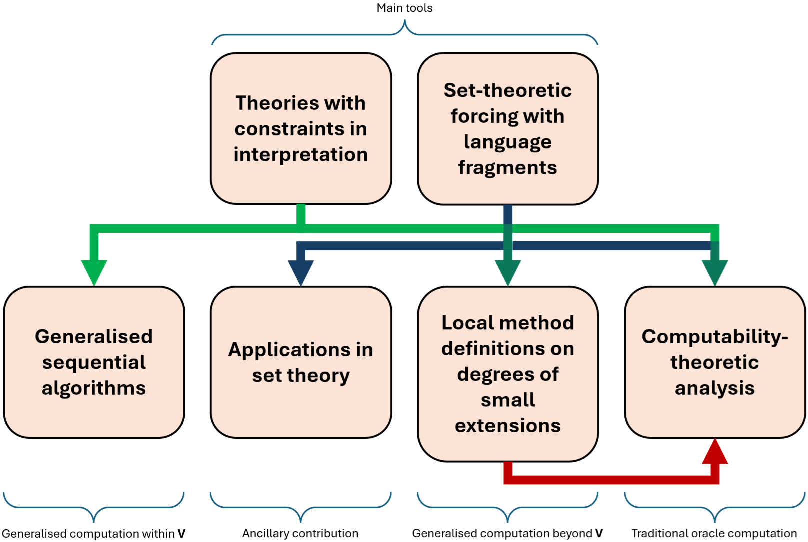
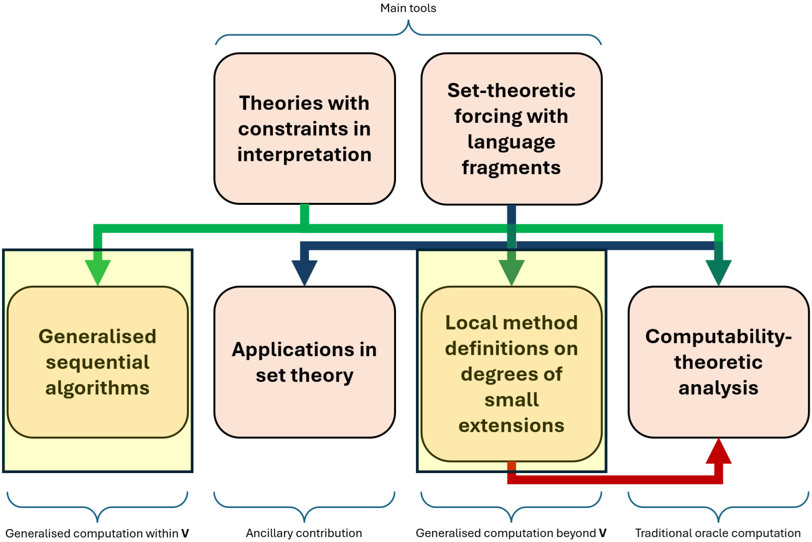
Overview Chart



Today's Focus



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Higher-Level Descriptions

- The explicit models of computation mentioned so far (BSSMs, ITTMs, etc.) are rather low-level and relatively concrete.
- Computer scientists were interested in more inclusive descriptions of algorithms.
- We want to be able to describe programs in pseudocode, instead of being bound to a fixed programming language.
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- Gurevich calls any such representation an *algorithm*.
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Gurevich's formulation involves proper classes, but it can be adapted to avoid size issues.

Definition

- A *sequential algorithm* comprises a set of *states*, including *initial states*, and a *transition function* mapping one state to another.
- Every state in a sequential algorithm is a first-order structure with the same base set and finite signature.
- In identifying the next state, the transition function determines changes to the current state based on finite information about the current state (*bounded exploration*).

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Generality and Intuitiveness

- States are analogous to Turing machine configurations.
- An initial state corresponds to a starting configuration upon receiving an input.
- A final state corresponds to an end configuration from which the output can be read.
- A transition function parallels that of a Turing machine: it codes how the machine behaves.
- Incidentally, this general framework of

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To the Infinite

- For an abstract computation model, sequential algorithms are simple to describe and very inclusive.
- It seems like the go-to candidate for generalisation, if we want an flexible and high-level model of infinitary computation.
- Two principles, isolated by Sieg in [8] as integral to classical models of computation, guide our search for the "correct" infinitary version of sequential algorithms.
  - *Boundedness*: "there is a fixed finite bound on the number of configurations a computor can immediately recognise".
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Redefining a Transition Function

- We want to talk about local features between two states.
- Since they are first-order structures with the same signature and base set, local features are synonymous with properties given through the truth predicate $\models$.
- We modify the predicate into $\models_2$ so that we can talk about any two such structures simultaneously.
- Every *generalised sequential algorithm* (*GSeqA*) has its transitions governed by a transition formula: transition from s_1 to s_2 iff $(s_1, s_2) \models_2 \phi$ for transition formula ϕ.

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Ordinal Base Sets

- To shift the burden of programming to the definition of the transition formula, we streamline the definitions of other components of a GSeqA.
- We specify the base set of any state to be a limit ordinal.
- More precisely, every GSeqA $\mathfrak{M}$ has an associated limit ordinal $\kappa(\mathfrak{M})$.
- Every state of a GSeqA $\mathfrak{M}$ is an expansion of the structure $(\kappa(\mathfrak{M}); \in)$ over a fixed signature $\sigma(\mathfrak{M})$ associated with $\mathfrak{M}$.
- This simplification is inspired by the fact that every set can be coded as a set of ordinals.
- The base set being an ordinal also fits the basic intuition of sequential memory when one thinks about computation.

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- To shift the burden of programming to the definition of the transition formula, we streamline the definitions of other components of a GSeqA.
- We specify the base set of any state to be a limit ordinal.
- More precisely, every GSeqA $\mathfrak{M}$ has an associated limit ordinal $\kappa(\mathfrak{M})$.
- Every state of a GSeqA $\mathfrak{M}$ is an expansion of the structure $(\kappa(\mathfrak{M}); \in)$ over a fixed signature $\sigma(\mathfrak{M})$ associated with $\mathfrak{M}$.
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Ordinal Parameters

- For technical purposes, we focus on the subclass of GSeqAs with ordinal parameters, called *generalised sequential algorithm with parameters* (*GSeqAP*s).
- Every GSeqAP $\mathfrak{M}$ has an associated finite set $\mathrm{p}(\mathfrak{M}) \subset \kappa(\mathfrak{M})$.
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- The interpretation of "computing B from A" is implicit in a sequential algorithm.
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Transition Formulas

- Let $\mathfrak{M}$ be a GSeqAP.
- For each declared variable $\ulcorner X \urcorner \in \sigma(\mathfrak{M})$, we associate it with a Δ_0 formula $\phi^{\ulcorner X \urcorner}$ of some specific form over $\sigma(\mathfrak{M})$.
- $\phi^{\ulcorner X \urcorner}$ specifies "pointwise" the next state's interpretation of $\ulcorner X \urcorner$ based on the current state's interpretation.
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Initial States

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- For each declared variable $\ulcorner X \urcorner \in \sigma(\mathfrak{M})$, we associate it with a Δ_0 formula $\psi^{\ulcorner X \urcorner}$ of some specific form over $\in$.
- $\psi^{\ulcorner X \urcorner}$ ought to specify "pointwise" the unique starting behaviour of $\ulcorner X \urcorner$.
- Then $\phi^d_{\mathfrak{M}}$ of $\mathfrak{M}$, defined to be the conjunction of the $\psi^{\ulcorner X \urcorner}$s, determines the starting behaviour of all declared variables.
- If $A \subset \kappa(\mathfrak{M})$, then there is a unique initial state s of $\mathfrak{M}$ such that s interprets In as A, Out as $\emptyset$ and $s \models \phi^d_{\mathfrak{M}}$ of $\mathfrak{M}$.
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Formal Definitions

Definition (L.)

A GSeqAP $\mathfrak{M}$ is a tuple

$$(\kappa(\mathfrak{M}), \sigma(\mathfrak{M}), \mathrm{p}(\mathfrak{M}), \phi^{\tau}_{\mathfrak{M}}, \phi^{d}_{\mathfrak{M}})$$

satisfying the properties we have mentioned.

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Let $S(\mathfrak{M})$ denote the set of states of $\mathfrak{M}$, derivable from $\kappa(\mathfrak{M})$, $\sigma(\mathfrak{M})$ and $\mathrm{p}(\mathfrak{M})$.

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$$\tau_{\mathfrak{M}} := \{(s_0, s_1) \in S(\mathfrak{M})^2 : \exists! x \in S(\mathfrak{M}) \ ((s_0, x) \models_2 \phi^{\tau}_{\mathfrak{M}}) \wedge (s_0, s_1) \models_2 \phi^{\tau}_{\mathfrak{M}}\}.$$

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Outline

A Call for Transfinite Runs

- *Runs* (or *computations*) are presumed to be finite in a sequential algorithm.
- Even with our restrictions, one can easily compress a finite run into a single step.
- If we allow transfinite inputs and outputs, it makes sense to allow transfinite runs.
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Taking Limits

- We want the definition of a limit state to depend locally on the states that come before it in a run.
- This can again be formalised using the first-order truth predicate in a "pointwise" Δ_0 way.
- Under this constraint, we show that with enough space (increasing $\kappa(\mathfrak{M})$ if necessary), taking limit/limsup/liminf wherever possible "pointwise" can simulate all other definitions of a limit state.

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Terminating Runs

Definition (L.)

A *terminating run* of a GSeqAP $\mathfrak{M}$ is a function from some successor ordinal δ into $S(\mathfrak{M})$ such that

- $Sq(0)$ is an initial state,
- $Sq(\alpha+1) = \tau_{\mathfrak{M}}(Sq(\alpha)) \neq Sq(\alpha)$ for all $\alpha < \delta - 1$,
- $Sq(\gamma)$ is obtained from $Sq \upharpoonright \gamma$ "pointwise" by taking liminf if possible and resetting to 0 otherwise, for limit ordinals $\gamma < \delta$, and
- $\tau_{\mathfrak{M}}(Sq(\delta - 1)) = Sq(\delta - 1)$.

We call δ the *length* of Sq.

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Relative Computability Relations

Definition (L.)

- A GSeqAP *computes B from A* iff it has a run that starts with In interpreted as A and terminates with Out interpreted as B.
- $B \leq^P_\kappa A$ iff some GSeqAP $\mathfrak{M}$ with $\kappa(\mathfrak{M}) = \kappa$ computes B from A.
- $B \leq^{P,s}_\kappa A$ iff some GSeqAP $\mathfrak{M}$ with $\kappa(\mathfrak{M}) = \kappa$ computes B from A in a terminating run of length $< \kappa$.
- $B \leq^P A$ iff $B \leq^P_\kappa A$ for some limit ordinal κ.

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Let A, B be sets of ordinals. Then $B \leq^P A$ iff $B \in L[A]$.

- This means the degrees of relative computability derived from $\leq^P$ are exactly the degrees of constructibility.
- The very restricted (at first glance) programming we allow in GSeqAPs actually has the full power of transfinite recursion.
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The Relations $\leq_\kappa^{P,s}$ and $\leq_\kappa^P$

- An ordinal κ is *admissible* iff $(L_\kappa;\in)$ is a model of Kripke-Platek set theory.
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If κ_1 and κ_2 are admissible, $\kappa_1 < \kappa_2$ and $B \leq_{\kappa_1}^P A$ (resp. $B \leq_{\kappa_1}^{P,s} A$), then $B \leq_{\kappa_2}^P A$ (resp. $B \leq_{\kappa_2}^{P,s} A$).

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In the following table we compare various relative computability relations when they are restricted to subsets of admissible ordinals α.

Relation / Property	$\leq_\alpha$	$\preceq_\alpha$	$\leq_\alpha^P$	$\leq_\alpha^{P,s}$
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- What do we mean by "sets outside V"?
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- Look at those sets that can be adjoined to V to give a nice CTM.
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Outer Models

- Let U_1 and U_2 be CTMs. U_2 is an outer model of U_1 iff
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- The binary relation "being an outer model of" is transitive.

Definition

Let V be a CTM. The *outward multiverse centred at V* is the set

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Due to Jensen's result on "coding the universe", we have the following theorem.

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Given a CTM V, $(\mathbf{M}_S(V), \subset)$ is a cofinal subposet of $(\mathbf{M}(V), \subset)$.

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The next proposition is easy to see.

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so "small extension(s)" and "degree(s) of small extensions" are often used interchangeably.

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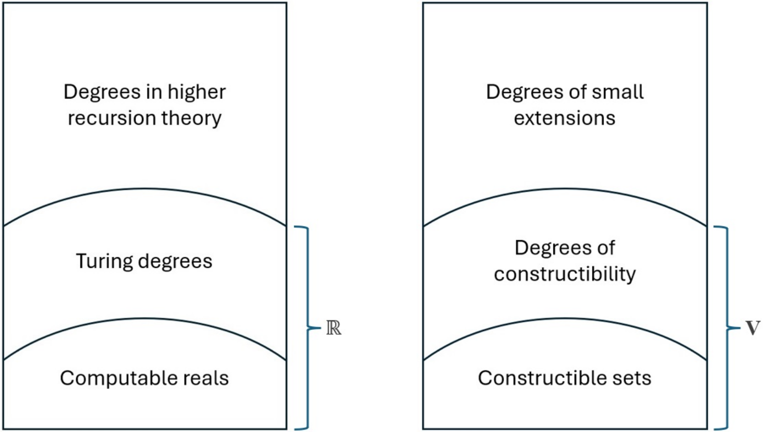

Figure: comparison between conventional notions of relative computability (left) and our generalised notions (right).

Multiverse of Generic Extensions

Definition

Let V be a CTM. The *outward generic multiverse centred at V* is the set

$$\mathbf{M}_F(V) := \{W : W \text{ is a forcing extension of } V\}.$$

The following fact is basic.

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Every generic extension is a small extension. Equivalently,

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Outline

Referring to Small Extensions in V

- Often it is useful to refer to small extensions of V within V.
- Such references in general cannot isolate any non-trivial small extension.
- Think of them as descriptions in V picking out sets of small extensions of V, when evaluated outside V.
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- *Theories with constraints in Interpretation* (*TCIs*), and their models, are a formalisation of this idea.
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Theories with Constraints in Interpretation

- TCIs provide bounds to the interpretation of a theory T over a signature σ.
- A TCI $\mathscr{T}$ expands on T by specifying additional requirements on a possible model $\mathfrak{M}$ of T.
- These requirements include
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Consistency of TCIs

Definition (L.)

A TCI is *consistent* iff it has a model in some outer model of V.

By examining a suitable forcing notion in V, we can decide if a TCI in V is consistent.

Proposition

Let $\mathscr{T}$ be a TCI. Then for all sufficiently large cardinals λ,

$$\Vdash_{Col(\omega,\lambda)} \exists \mathfrak{M} \text{ (``}\mathfrak{M} \textit{ is a model of } \mathscr{T}\text{'')} \iff \mathscr{T} \textit{ is consistent.}$$

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Local Method Definitions

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A *local method definition of* V is a non-empty class of TCIs definable in V.

- In the meta-theory, define a function Eval^V from the set of TCIs $\mathscr{T} \in V$ into the set of small extensions of V, such that

$$\mathrm{Eval}^V(\mathscr{T}) = \{V[\mathfrak{M}] \in \mathbf{M}_S(V) : \mathfrak{M} \text{ is a model of } \mathscr{T}\}.$$

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Comparing Local Methods

Definition (L.)

If X and Y are local method definitions of V, $X \leq^M Y$ denotes the statement

> "there is a function $F : X \longrightarrow Y$ definable in V such that $\emptyset \neq \text{Eval}^V(F(\mathscr{T})) \subset \text{Eval}^V(\mathscr{T})$ for all consistent $\mathscr{T} \in X$".

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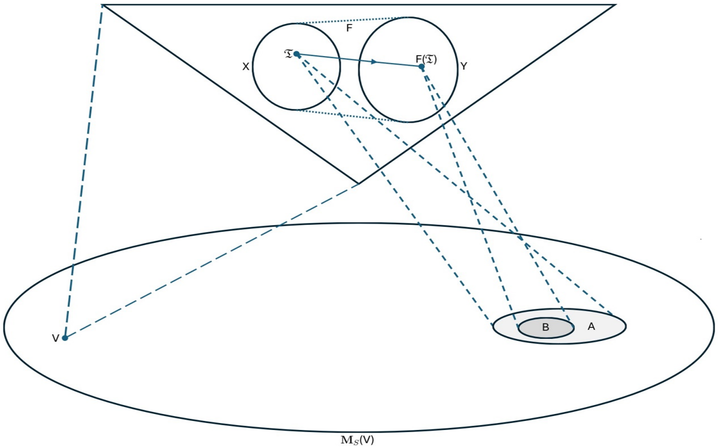

Figure: Visual representation of a function F witnessing $X \leq^M Y$, where X and Y are local method definitions.

The Local Method Hierarchy

- We can also define the complexity of a TCI by just looking at its first-order theory.
- For example, a Σ_n TCI has a first-order theory containing only Σ_n sentences.
- The classes of Σ_n and Π_n TCIs — denoted $\Sigma^{\mathsf{M}}_{\mathsf{n}}$ and $\Pi^{\mathsf{M}}_{\mathsf{n}}$ respectively — then form a hierarchy under the relation $\leq^M$, called the local method hierarchy.

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Theorem (L.)

Let $1 \leq n < \omega$. Then for every $\mathscr{T} \in \Pi^{\mathsf{M}}_{\mathsf{n}+1}$ there is $\mathscr{T}' \in \Sigma^{\mathsf{M}}_{\mathsf{n}}$ such that

$$\mathrm{Eval}^V(\mathscr{T}) = \mathrm{Eval}^V(\mathscr{T}').$$

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$\Pi^{\mathsf{M}}_{\mathsf{n+1}} \leq^M \Sigma^{\mathsf{M}}_{\mathsf{n}}$ *for all $1 \leq n < \omega$.*

The Local Method Hierarchy

Theorem (L.)

Let $1 \leq n < \omega$. Then for every $\mathscr{T} \in \Pi^{\mathsf{M}}_{\mathsf{n+1}}$ there is $\mathscr{T}' \in \Sigma^{\mathsf{M}}_{\mathsf{n}}$ such that

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$\Pi^{\mathsf{M}}_{\mathsf{n+1}} \leq^M \Sigma^{\mathsf{M}}_{\mathsf{n}}$ *for all* $1 \leq n < \omega$.

Set Forcing

- There is an obvious and uniform way of representing any forcing notion $\mathbb{P}$ as a Π_2 TCI $\mathscr{T}(\mathbb{P})$.
- Thus set forcing, as a technique of accessing small extensions of V, can be represented by the local method definition

$$\mathrm{Fg} := \{\mathscr{T}(\mathbb{P}) : \mathbb{P} \text{ is a forcing notion}\}.$$

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Set Forcing is Σ_1 (is Π_2)

Theorem (L.)

$\mathrm{Fg} \equiv^M \Sigma_1^{\mathrm{M}}$.

Proof Idea.

By adapting our framework for constructing forcing notions to the context of TCIs, we associate with each Π_2 TCI $\mathscr{T}$ a forcing notion $\mathbb{P}(\mathscr{T})$, such that a unique model of $\mathscr{T}$ can be read off every $\mathbb{P}(\mathscr{T})$-generic filter over V.

This allows us to define a class function witnessing $\Pi_2^{\mathrm{M}} \leq^M \mathrm{Fg}$. $\square$

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A Complementary Theorem

Theorem (L.)

Let $\emptyset \neq X \subset \mathbf{M}_S(V)$ be defined by

$$X = \{V[x] : \text{“}x \text{ is a } Y\text{-generator”}\},$$

where “X is a Y-generator” has a sufficiently absolute definition. Moreover, assume there is a set $z \in V$ such that

$$\emptyset \neq \{a \subset z : \text{“}a \text{ is a } Z\text{-code of } x\text{”}\}$$

has a definition both uniform in Y-generators x and absolute for outer models of V. Then there is a forcing notion $\mathbb{P}$ for which

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- The hypotheses of the previous theorem are a conservative attempt at formalising what a definition of a class of small extensions of V through their generators should entail, if the said definition depends locally on bounded amount of information in V.
- As a result, the theorem can be interpreted thus: forcing is the most powerful means among all such means of defining classes of small extensions of V.

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Let $n < \omega$, $\vec{y} \in V^n$, and φ be a $(n+1)$-ary Δ_1 formula in the Lévy hierarchy. Define

$$U(\vec{y}, \varphi) := \{V[x] : \exists x' \in V \ (x \subset x') \text{ and } \varphi(\vec{y}, x)\}.$$

If $U(\vec{y}, \varphi) \neq \emptyset$, then there is a forcing notion $\mathbb{P} \in V$ for which $\mathrm{Eval}^V(\mathfrak{T}(\mathbb{P})) \subset U(\vec{y}, \varphi)$.

- Known methods of generating small extensions of a ground model are Δ_1-definable with the ground model and its members as parameters.
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Further Results

- We can refine the function witnessing $\Pi_2^{\mathrm{M}} \leq^M \mathrm{Fg}$ by iteratively pruning the $\mathbb{P}(\mathscr{T})$s of atoms.
    - This pruning process is analogous to taking Cantor-Bendixson derivatives.
- There are analogues of the previous theorem in V pertaining to certain countable Π_2 TCIs.
    - Essentially, we can define functions F of low complexity taking these TCIs to reals such that every $F(\mathscr{T})$-1-generic real codes a model of $\mathscr{T}$.

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Some Open Questions

(Q1) What structural properties of $(\mathbf{M}_F(L), \subset)$ also hold for $(\mathbf{M}_S(L), \subset)$?

(Q2) For example, is $(\mathbf{M}_S(L), \subset)$ downward directed?

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Thank You!